

SmartShelf

Smart Retail Inventory & First-Expiry-First-Out (FEFO) Management System

TEAM REPORT - PART 3 of 4 | Team Member 3: [Your Name]

Your role on the project: Intelligence & analytics owner

What YOU present and defend: Spoilage risk scoring, dynamic pricing advice, the promotion approval flow, analytics and forecasting, alerts/notifications and waste recovery.

How to use this part

Read your sections below until you can walk someone through them without notes. Then practise the Q&A cues at the end - they mirror what a panel typically asks. If you do not know a detail outside your part, say so and hand it to the right teammate.

You must be able to explain

- How the 0-100 risk score is computed (inputs, levels).
- How risk maps to a recommended discount percent.
- The promotion lifecycle and who approves what.
- The dashboard KPIs, velocity, demand and forecast.
- How alerts and the unread badge work.
- What waste recording measures (loss vs recovery).

1. Spoilage risk engine

- Every live batch is scored 0-100 from: days to expiry, product types, sales velocity (units/day over 30 days) and stock on hand.
- Levels: LOW (0-20), MEDIUM (21-45), HIGH (46-70), CRITICAL (71-100); CRITICAL means imminent loss.
- The same score feeds pricing, the dashboard's risk distribution and the top-risk list.
- Read it via GET /api/ai/risk (all batches, sorted) or /api/ai/risk/ (one batch).

2. Dynamic pricing advice

The pricing engine converts a risk score into a recommended discount percent (LOW 0-10%, MEDIUM 10-20%, HIGH 20-40%, CRITICAL 40-60%) and explains it in plain language - e.g. 'expires in 5 days'. GET /api/ai/pricing lists every at-risk batch with unit price, the recommended percent, the discounted price and the reason.

3. Promotion approval workflow

- Manager creates a promotion as DRAFT (discount type PERCENTAGE/FIXED, value, optional product, category or minimum quantity).
- Manager clicks SUBMIT -> PENDING_APPROVAL.
- Only ADMIN can APPROVE (to ACTIVE) or REJECT.
- Manager can later toggle ACTIVE/INACTIVE; the state machine is enforced at the API level.
- The system advises, the manager proposes, the admin disposes - a deliberate control step.

4. Analytics and forecasting

- Dashboard KPIs: product/stock/batch counts, inventory value, revenue, sales count, waste value, at-risk items, expiring/expired counts.
- Also shown: expiry summary, risk distribution (LOW/MEDIUM/HIGH/CRITICAL), daily sales trend, top-10 risk list and low-stock list.
- Velocity classifies products by 30-day sales rate; demand-analysis returns per-product units/day.
- Forecast projects the next N days (default 7) from ~60 days of history with a simple model.

5. Alerts and notifications

- Expiry scan classifies batches (OK / EXPIRING_SOON / EXPIRED) within a 30-day window.
- Three generators: low stock (qty <= reorder level), expiry, and high risk.
- Notifications are per-user; unread-count drives the navbar badge; read-all clears them.

6. Waste and revenue recovery

- Staff record spoilage/damage: product, quantity, reason (EXPIRED/DAMAGED/SPOILED/OTHER), optional batch/location.
- The engine deducts the exact batch and stores unit cost, potential loss and a recovery estimate.
- Summary aggregates losses and recovery by reason and period.

7. Endpoints you should be able to quote

METHOD	PATH	ACCESS	PURPOSE
GET	/api/ai/risk	Any user	Risk score for every live batch
GET	/api/ai/risk/	Any user	Single batch score
GET	/api/ai/pricing	Any user	Recommended discounts + reasons
POST	/api/promotions//submit	MANAGER+	Submit for approval
POST	/api/promotions//approve	ADMIN	Approve -> ACTIVE
GET	/api/dashboard	Any user	KPI suite + trends + risk
POST	/api/alerts/generate	MANAGER+	Low stock / expiry / high risk
POST	/api/wastage	STAFF+	Record loss + recovery

8. Likely questions and answer cues

Q. How is the 0-100 risk score computed?

A. From batch-specific inputs - days to expiry, product perishability, sales velocity over 30 days and stock on hand - blended into one score with levels LOW/MEDIUM/HIGH/CRITICAL.

Q. What does CRITICAL mean and what happens then?

A. 71-100: expiry within days. The app flags the batch, suggests a strong discount (40-60%) and surfaces it in the risk list and dashboard.

Q. Why does a discount need approval?

A. Discounting is a business decision, not a math one. Managers propose (DRAFT -> SUBMIT) and admins approve/reject so money is never auto-discounted without a human.

Q. How does the forecast work?

A. It builds a per-product demand trend from roughly 60 days of sales and projects the next N days; the model is deliberately simple and honest.

Q. What KPIs are on the dashboard?

A. Stock units and value, batch count, revenue and sales count, waste value, at-risk items, expiring/expired counts plus a daily sales trend and low-stock list.

Q. How do alerts get to staff?

A. Scans/generators write per-user notifications; the navbar unread badge shows the count and each notification can be marked read.

Q. What does waste recording measure?

A. Quantity, reason, unit cost and potential loss, plus an estimated recovered revenue/savings so the business sees the true cost of expiry.